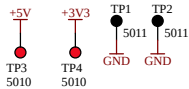
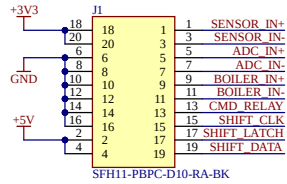


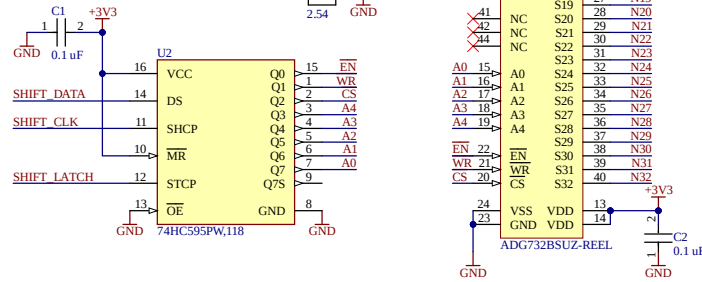
BOARD TO BOARD CONNECTOR



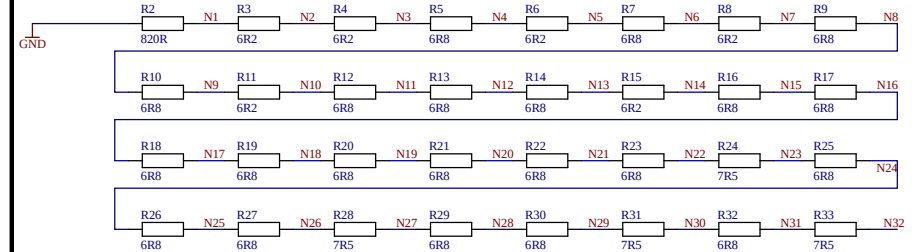
MUX CMD SIMULATION

$$I_{dyn} = CPD \cdot VCC - I_i + \Sigma (CL \cdot VCC \cdot f_o) \\ = 115e-12 \cdot 3.3 - 1.0e6 + 40e-12 \cdot 3.3 \cdot (1.0e6/8) \\ = 0.3795e-3 + 0.0165e-3 \\ = 0.3960e-3[A] = 0.396[mA] \text{ par composant}$$

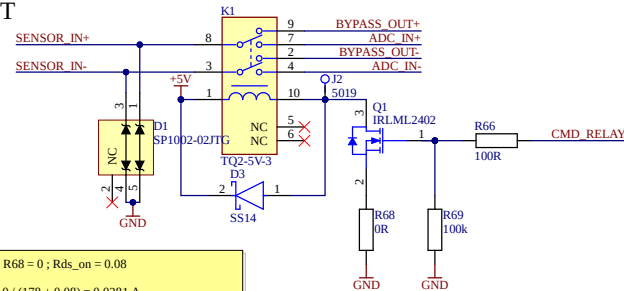
$$I_{total_max_par_composant} = I_{dyn} + ICC_max = 0.396[mA] + 0.080[mA] = 0.476[mA]$$



RESISTOR NETWORK SMD



BYPASS INPUT



Data: $V = 5.0$; $R_{coil} = 178$; $R_{68} = 0$; $R_{ds_on} = 0.08$

$$I = V / (R_{coil} + R_{ds_on}) = 5.0 / (178 + 0.08) = 0.0281 \text{ A}$$

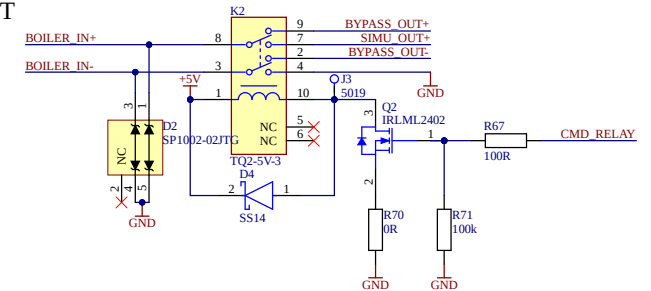
$$P_{total_5V} = V \cdot I = 5.0 \cdot 0.0281 = 0.1405 \text{ W}$$

$$P_{coil} = I^2 \cdot R_{coil} = 0.0281^2 \cdot 178 = 0.1404 \text{ W}$$

$$P_{MOSFET} = I^2 \cdot R_{ds_on} = 0.0281^2 \cdot 0.08 = 0.000063 \text{ W}$$

OFF state: $I = 0 \text{ A}$; $P = 0 \text{ W}$

BYPASS OUTPUT



ObloSolution * * * *	Mod.	a		Dessiné	LMS	19.08.2025	A3		Echelle :
		b		Contrôlé	*				_____
		c							
		d		Remplace					
	Nom du projet						No		
	RegulationChauffage						2510B		

Line #	Name	Description	Designator	Revision ID	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	0.1 uF	CAP CER 0.1 uF 10V X5R 0805	C1, C2			2	Yageo Group	CC0805KRX7R9BB104	Volume Production	Farnell	3019949	0.03003	0.30028
	SP1002-02JTG	16V Pince 2 A (8/20µs) Ipp TVS - Diodes Montage en surface SC-70- 5	D1, D2			2	Littelfuse	SP1002-02JTG	Volume Production	DigiKey	F4238CT-ND	0.37	0.74
	SS14	DIODE SCHOTTKY 40V 1A SMA	D3, D4			2	onsemi	SS14	Volume Production	Farnell	1467537	0.26592	1.33
	SFH11-PBPC-D10-RA-BK	No Description Available	J1			1	Sullins	SFH11-PBPC-D10-RA-BK	Volume Production	Digikey	S9205-ND	1.16	1.16
	5019		J2, J3			2	Keystone	5019	Volume Production	DigiKey	36-5019CT-ND	0.37	0.74
	TQ2-5V-3	RELAY TELECOM DPDT 1A 5VDC	K1, K2			2	Panasonic	TQ2-5V-3	Volume Production	DigiKey	255-2257-ND	3.48	6.96
	PCB		PCB1			1	Eurocircuit	-	Manual Solution	Eurocircuit	-	69.37	69.37
	IRLML2402		Q1, Q2			2	Infineon	IRLML2402TRPBF	Obsolete	Mouser	942-IRLML2402TRPBF	0.333	0.666
	200R	200 Ohms 0.5W, 1/2W PC Pins Through Hole Trimmer Potentiometer Cermet 1.0 Turn Side Adjustment	R1			1	Bourns	3386H-1-201LF	Volume Production	Farnell	9354930	1.78	1.78
	820R	820 Ohms ±1% 0,125W, 1/8W Pavé résistif 0805 (2012 métrique) Couche épaisse - RC0805FR- 13820RL	R2			1	Yageo Group	RC0805FR-13820RL	Volume Production	DigiKey	13-RC0805FR-13820RLCT- ND	0.1	0.1
	6R2	RES 6.2 OHM 1% 1/8W 0805 RC0805FR-076R2L	R3, R4, R6, R8, R11, R15			6	Yageo Group	RC0805FR-076R2L	Volume Production	Farnell	3952295	0.03663	0.36632
	6R8	RES 6.8 OHM 1% 1/8W 0805	R5, R7, R9, R10, R12, R13, R14, R16, R17, R18, R19, R20, R21, R22, R23, R25, R26, R27, R29, R30, R32			21	Yageo Group	RC0805FR-076R8L	Volume Production	DigiKey	311-6.80CRCT-ND	0.019	0.399
	7R5	RES SMD 7.5 OHM 1% 1/8W 0805 AC0805FR-077R5L	R24, R28, R31, R33			4	Yageo Group	AC0805FR-077R5L	Volume Production	DigiKey	13-AC0805FR-077R5LCT- ND	0.027	0.27
	100R	RES 100ohm ±1% 0.125W 2012 (0805 INCH)	R66, R67			2	Yageo Group	RC0805FR-07100RL	Volume Production	Farnell	9237364	0.01656	0.16561
	0R	RES 0ohm ±1% 0.125W 2012 (0805 INCH)	R68, R70			2	Vishay	CRCW08050000Z0EA	Volume Production	DigiKey	541-0.0ACT-ND	0.10496	0.20993
	100k	RES 100kohm ±1% 0.125W 2012 (0805 INCH)	R69, R71			2	Yageo Group	RC0805FR-07100KL	Volume Production	Farnell	9237879	0.01551	0.15511
	5011	Test Point, Black, Through Hole, RoHS, Bulk	TP1, TP2			2	Keystone	5011	Volume Production	Farnell	2431148	0.14443	0.28887
	5010	Test Point, Red, Through Hole, RoHS, Bulk	TP3, TP4			2	Keystone	5010	Volume Production	DigiKey	36-5010-ND	0.39	0.78
	ADG732BSUZ-REEL	1.8 to 5.5 V / +/- 2.5 V, 4 Ohm, 32- Channel Analog Multiplexer, -40 to 85 degC, 48-pin QFP (SU-48), Reel	U1			1	Analog Devices	ADG732BSUZ-REEL	Volume Production	Farnell	4032557	17.29	17.29
	74HC595PW,118	IC SHIFT REGISTER 8BIT 16TSSOP	U2			1	Nexperia	74HC595PW,118	Volume Production	DigiKey	1727-3068-1-ND	0.28	0.28
	2.54	PIN HEADER 2.54 1x10	X1			1	Würth Elektronik	61300411121		DigiKey	732-5317-ND	0.17494	0.17494